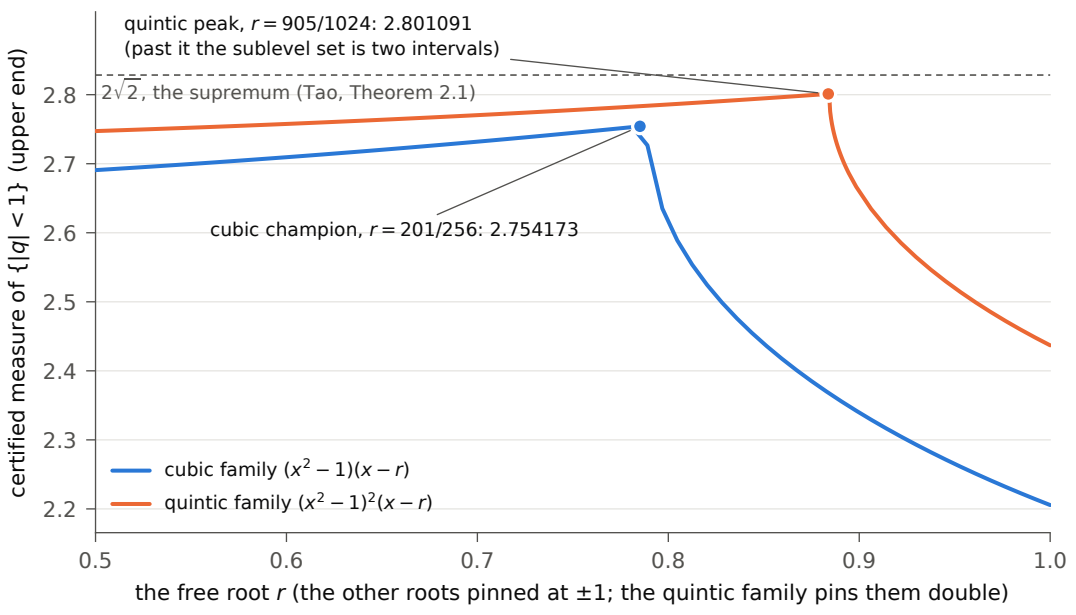
quintic peak, $r = 905/1024$: 2.801091
(past it the sublevel set is two intervals)
$2\sqrt{2}$, the supremum (Tao, Theorem 2.1)
cubic champion, $r = 201/256$: 2.754173
cubic family $(x^2-1)(x-r)$
quintic family $(x^2-1)^2(x-r)$
certified measure of $\{|q|<1\}$ (upper end)
2.8
2.7
2.6
2.5
2.4
2.3
2.2
0.5
0.6
0.7
0.8
0.9
1.0
the free root $r$ (the other roots pinned at $\pm 1$; the quintic family pins them double)